

Q1.

A train is travelling at maximum speed with its engine using its maximum power of 1800 kW

When travelling at this speed the train experiences a total resistive force of 40 000 N

Find the maximum speed of the train.

Circle your answer.

22 m s⁻¹

45 m s⁻¹

54 m s⁻¹

90 m s⁻¹

(Total 1 mark)

Q2.

A car, of mass 1200 kg, moves on a straight horizontal road where it has a maximum speed of 40 m s⁻¹

When the car travels at a speed of v m s⁻¹ it experiences a resistance force which can be modelled as being of magnitude $30v$ newtons.

- (a) Show that the power output of the car is 48 000 W, when it is travelling at its maximum speed.

(3)

- (b) Find the maximum acceleration of the car when it is travelling at a speed of 25 m s⁻¹

(4)

(Total 7 marks)

Q3.

A train, of mass 22 tonnes, moves along a straight horizontal track. A constant resistance force of 5000 N acts on the train. The power output of the engine of the train is 240 kW.

Find the acceleration of the train when its speed is 20 m s⁻¹.

(Total 6 marks)

Q4.

A car travels along a straight horizontal road. When its speed is v m s⁻¹, the car experiences a resistance force of magnitude $25v$ newtons.

- (a) The car has a maximum constant speed of 42 m s⁻¹ on this road.

Show that the power being used to propel the car at this speed is 44 100 watts.

(2)

- (b) The car has mass 1500 kg.

Find the acceleration of the car when it is travelling at 15 m s⁻¹ on this road under a power of 44 100 watts.

(4)

(Total 6 marks)

Q5.

A pump is being used to empty a flooded basement.

In one minute, 400 litres of water are pumped out of the basement.

The water is raised 8 metres and is ejected through a pipe at a speed of 2 m s^{-1} .

The mass of 400 litres of water is 400 kg.

- (a) Calculate the gain in potential energy of the 400 litres of water.

(1)

- (b) Calculate the gain in kinetic energy of the 400 litres of water.

(1)

- (c) Hence calculate the power of the pump, giving your answer in watts.

(2)

(Total 4 marks)

Q6.

A train consists of an engine and five carriages. A constant resistance force of 3000 N acts on the engine, and a constant resistance force of 400 N acts on each of the five carriages.

The maximum speed of the train on a horizontal track is 90 km h^{-1} .

- (a) Show that this speed is 25 m s^{-1} .

(1)

- (b) Hence find the maximum power output of the engine. Give your answer in kilowatts.

(3)

(Total 4 marks)

Q7.

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A train, of mass 60 tonnes, travels on a straight horizontal track. It has a maximum speed of 40 m s^{-1} when its engine is working at 800 kW.

- (a) Find the magnitude of the resistance force acting on the train when the train is travelling at its maximum speed.

(3)

- (b) When the train is travelling at 40 m s^{-1} , the power is turned off. Assume that the resistance force is constant and is equal to that found in part (a). Also assume that this resistance force is the only horizontal force acting on the train.

Use an energy method to find how far the train travels when slowing from 40 m s^{-1} to 36 m s^{-1} .

(4)

(Total 7 marks)

Q8.

A car of mass 600 kg is driven along a straight horizontal road. The resistance to motion

of the car is kv^2 newtons, where $v \text{ m s}^{-1}$ is the velocity of the car at time t seconds and k is a constant.

- (a) When the engine of the car has power 8 kW, show that the equation of motion of the car is

$$600 \frac{dv}{dt} - \frac{8000}{v} + kv^2 = 0$$

(4)

- (b) When the velocity of the car is 20 m s^{-1} , the engine is turned off.

- (i) Show that the equation of motion of the car now becomes

$$600 \frac{dv}{dt} = -kv^2$$

(1)

- (ii) Find, in terms of k , the time taken for the velocity of the car to drop to 10 m s^{-1} .

(5)

(Total 10 marks)

Q9.

A motorcycle has a maximum power of 72 kilowatts. The motorcycle and its rider are travelling along a straight horizontal road. When they are moving at a speed of $V \text{ m s}^{-1}$, they experience a total resistance force of magnitude kV newtons, where k is a constant.

The maximum speed of the motorcycle and its rider is 60 m s^{-1} .

Show that $k = 20$.

(Total 3 marks)

Q10.

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A car of mass 1500 kg is travelling along a straight horizontal road. When the car is travelling at a speed of $v \text{ m s}^{-1}$, it experiences a resistance force of magnitude $35v$ newtons.

- (a) On this road, the car has a maximum speed of 50 m s^{-1} .

Show that the maximum power of the car is 87 500 watts.

(4)

- (b) Find the maximum possible acceleration of the car when its speed on the road is 30 m s^{-1} .

(5)

(Total 9 marks)